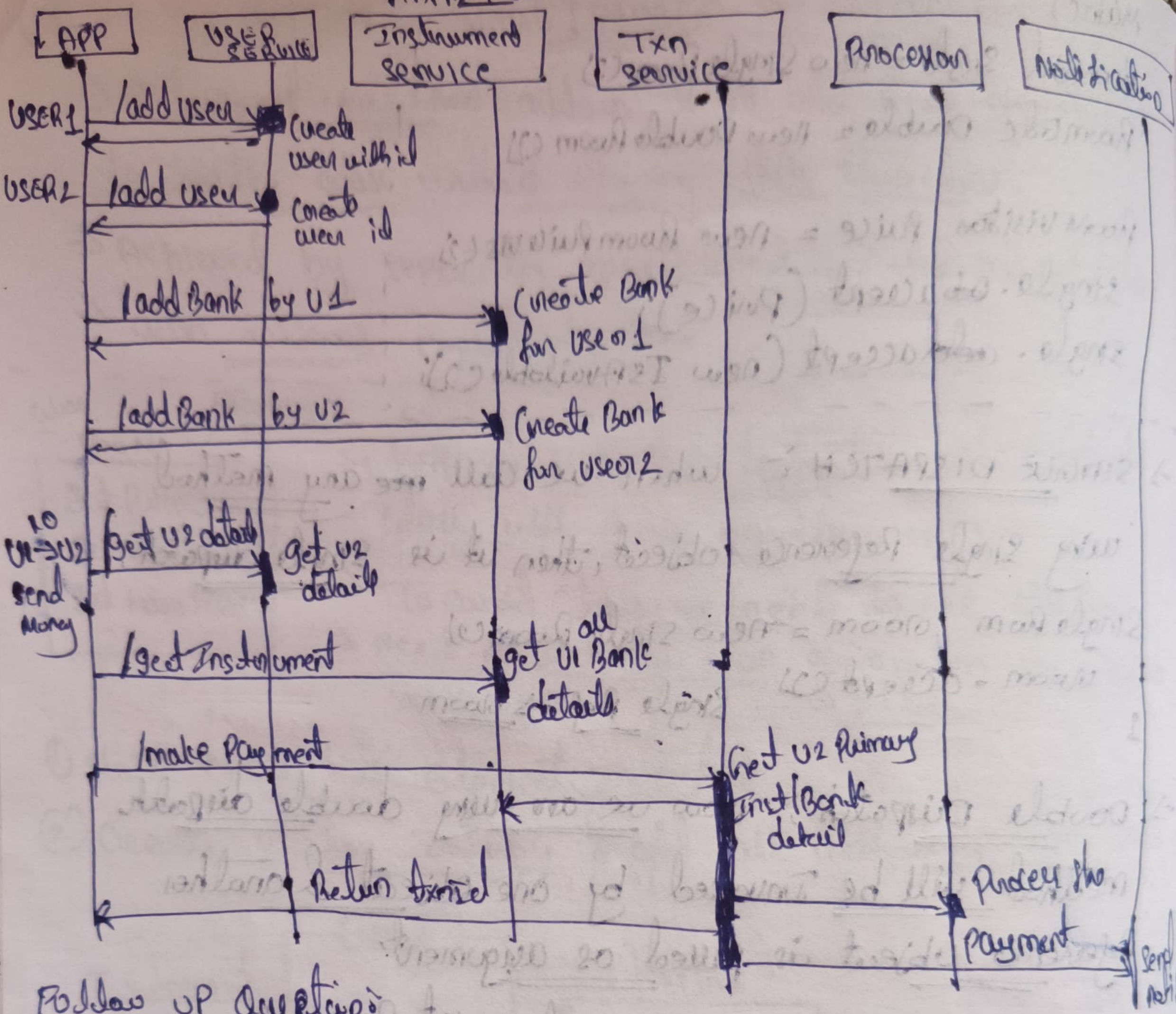
(20) PAYMENT GATEWAY :- Sending Money from U1 → U2

Flow

- (1) ADD | Update | Delete USER
- (2) ADD | Update | Delete Instrument
  - BANK
  - CARD
- (3) make Payment U1 → U2
  - → Search U2 , to send amount
    - email
    - Phone
  - → select amound & Instrument
  - → Pass information to Processor.
- (4) send notification for all operation
  - → Email
  - → sms
- (5) See all transaction history

HAPPY FLOW


APP
USER SERVICE
Instrument Service
Txn Service
Processor
Notification
USER1
/add user
Create user with id
USER2
/add user
Create user id
/add Bank by U1
Create Bank for user 1
/add Bank by U2
Create Bank for user2
U1→U2 send Money
/get U2 details
get U2 details
/get Instrument
get all U1 Bank details
/make Payment
Get U2 Primary Inst/Bank detail
Return txn id
Process the payment
Send noti

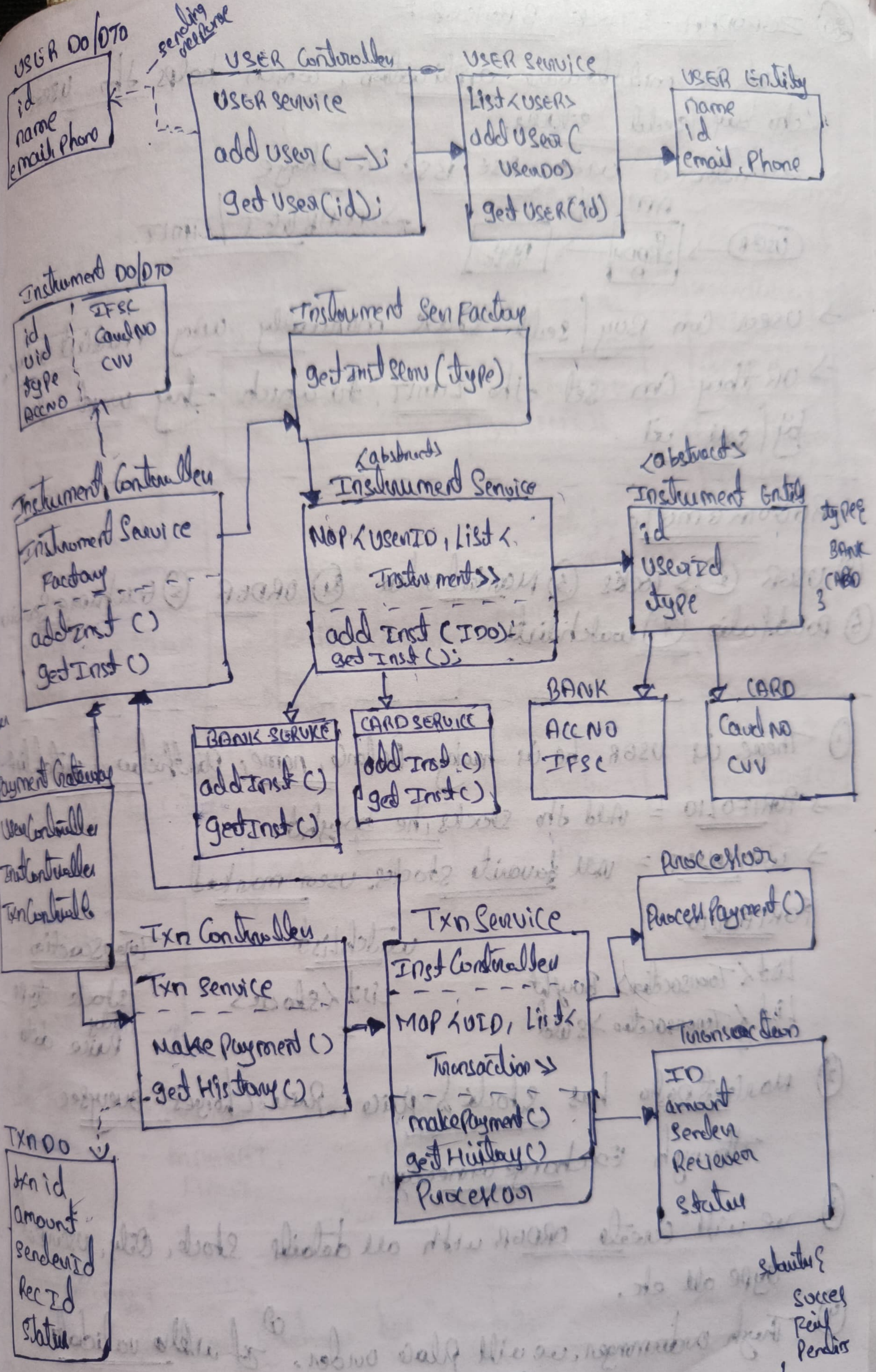
Follow up Questions:

① Payment processing can take upto 3-5 days. To complete payment.

-> we can make use of ASYNC operation :-

① Invoke Processor to validate :
  i. U1 has sufficient balance
  ii. U2 Bank is valid.

② Then save transaction as PENDING
  i. Invoke Process in ASYNC & complete PENDING txn
  ii. once it comes back, update the txn status.


USER DO/DTO
id
name
email, Phone
sending response
USER Controller
USER service
addUser(—);
getUser(id);
USER Service
List<USER>
addUser(
UserDO)
getUSER(id)
USER Entity
name
id
email, Phone
Instrument DO/DTO
id
uid
type
AccNO
IFSC
CardNO
CVV
Instrument Serv Factory
getInstServ(type)
<abstract>
Instrument Service
MAP<UserID, List<
Instrument>>
addInst(IDO);
getInst();
<abstract>
Instrument Entity
id
UserID
type
type{
BANK
CARD
}
Instrument Controller
Instrument Service
Factory
addInst()
getInst()
BANK SERVICE
addInst()
getInst()
CARD SERVICE
addInst()
getInst()
BANK
ACCNO
IFSC
CARD
CardNO
CVV
Payment Gateway
UserController
InstController
TxnController
Txn Controller
Txn service
makePayment()
getHistory()
Txn Service
Inst Controller
MAP<UID, List<
Transaction>>
makePayment()
getHistory()
Processor
Processor
ProcessPayment()
Transaction
ID
amount
Sender
Reciever
status
status{
Succes
Fail
Pending
}
TxnDO
txnid
amount
senderId
RecId
status